EliteDesign: Elite-Guided Learning for RNA Design via the Cross-Entropy Method

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Abstract

The aim of RNA design is to find, within an exponentially large search space, a nucleotide sequence that folds into a given target structure. Efficient RNA design could aid in the development of RNA-based technologies such as riboswitches and ribozymes. However, finding sequences that fold into the target structure with high probability remains challenging, particularly for long RNAs. To address this challenge, we introduce an approach that uses the cross-entropy method to iteratively refine a distribution over valid RNA sequences. Our method outperforms the state-of-the-art RNA design method, SamplingDesign, on Eterna100, achieving a better average target folding probability while being 500 times faster. It also scales to longer RNA structures, achieving 2.55 times the average target folding probability of SAMFEO on 10 16S rRNA targets ranging from 950 to 1,525 nucleotides, with a 15-fold reduction in average runtime per puzzle.